# Virtual University of Pakistan

# (Bif401)

# (Bioinformatics)

**Midterms Past Papers**

**Email address**

bif401@vu.edu.pk

**Course Instructor**

**(Dr. Safee Ullah Choudry)**

**Created By Sadaf Parvez**

**Group of VU Biologists**

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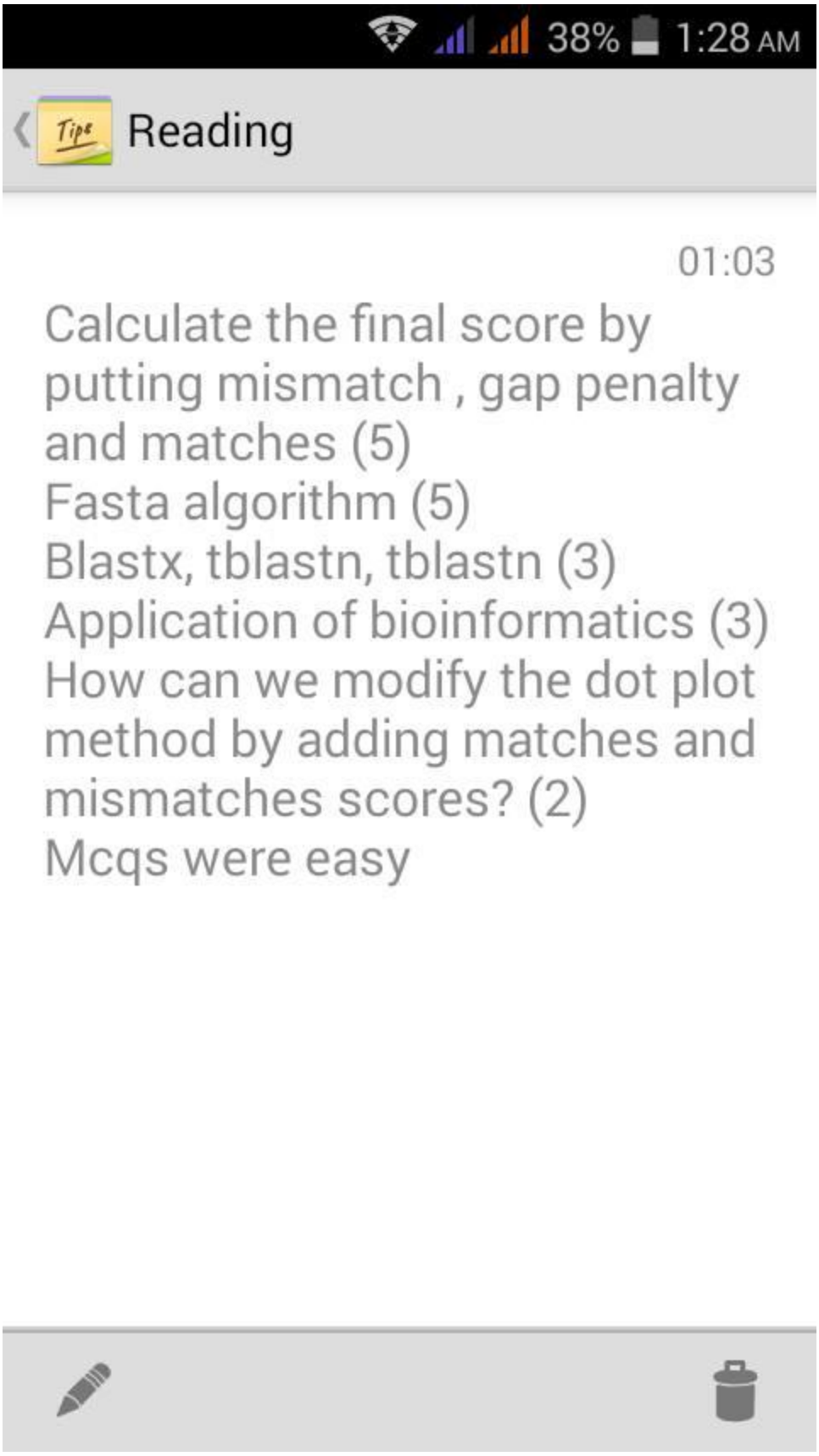
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(MID papers 2018)

Bif401
30-6-2018
10:30am
Qno.1.Dynamic Programming?
Qno.2.steps of BLOSUM matrix?
Qno.3.Application of bioinformatics?
Qno.4.hypothesis of wobble pair?
Qno.5.identity sequence and similarity sequence?also write the formula for identity sequence?
bif401
mcqs
definitions the
mRNA, psudoknot,node or ek glutamine ko kis letter sy represent karty (Q)
tissue culturing is a bioinformatic technique(no) or FASTA is the online software can perform genome proteom similarity search (true)
baqi b easy thy
what mean of substitution and indel in sequencing?2
BLAST stand for?2
MSA and PWA difference?3
objective of comparing sequence?3
pam matrix steps?5
local and global alignments?5
best of luck

>recal and discuss tabuler 3D protien in PDB (2)
>how biological simulation give benefit to society (2)
>write and describe the distance formula between 2 sequences (3)
>how bulges are formed (5)
>how Dp deals with the match mismtch nd gaps?(a sequnces was also given) (5)
only 2 que was repeated . mcqs was easy .

Bif401
Define mutation (2)
Databases use for proteins (2)
Generally used databases and which type of information they store (3)
Scoring matrix (5)
Types of RNA and their functions (5)




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My today paper of bif401.
1.difference b/w DNA and RNA ?
2.rooted and unrooted tree difference?
3.dynammic programming to create scoring matrix?r 1 series sequence dia hua tha scoring matrix maloom krny k liy?
4.pam matrix k steps?
5.uniprot and swiss prot ?
Bif401 today paper
What is mutation 2
Uses of clustalw 2
Pseudonots 3
Uniprot 3
Scoringatrix formula 5
Pam matric steps5
BIF401

Solved papers (plz correct if u see any mistake)

solved by Zareen Fatima

**Define mutation (2)**

Ans:

Mutations are:

• Replacement of amino acids with certain other amino acids in proteins

• Replacement of nucleotides with certain other nucleotides in DNA/RNA

**Databases use for proteins (2)**

Ans: uniport and swisport

**Generally used databases and which type of information they store (3)**

Ans:

Genebank. Pdb, uniport swisport

- Locus

• Accession number

- Sequence
- Molecular mass

• Authors

• Journal etc.

## Types of RNA and their functions (5)

Ans:

RNA can be divided into two categories

• Coding RNAs

• Non-Coding RNAs

• Coding RNAs as is obvious from their name, code for Proteins

• Non-Coding RNAs regulate/assist in the process of translation

Types of RNA

1. Messenger RNA (mRNA) Messenger RNA (mRNA) carries the genetic information copied from DNA in the form of a series of three-**base**code "words," each of which specifies a particular amino acid

2. Transfer RNA (tRNA) is the key to converting the code words in mRNA.

3. Ribosomal RNA (rRNA) associates with a set of proteins to form ribosomes. These complex structures, which physically move along an **mRNA** molecule, catalyze the assembly of **amino acids** into protein chains. They also bind tRNAs and various accessory molecules necessary for **protein synthesis**.

4. Micro RNAs (miRNA)

functions in RNA silencing and post-transcriptional regulation of gene expression.

5. Small Interfering RNA (siRNA)

Small (or short) interfering **RNA** (**siRNA**) is the most commonly used **RNA** interference (**RNAi**) tool for inducing short-term silencing of protein coding genes.**siRNA** is a synthetic **RNA** duplex designed to specifically target a particular mRNA for degradation.

## Uses of clustalw 2

Ans: CLUSTALW can:

• create multiple alignments,

• optimize existing alignments,

• profile analysis &

• create phylogenetic trees

## Pseudonots 3

Ans: • Tertiary or 3' structures may form pseudoknots

• A simple graph-based approach "Circular Plot" can help us detect the pseudoknots

A **pseudoknot** is a nucleic acid secondary structure containing at least two stem-loop structures in which half of one stem is intercalated between the two halves of another stem. Pseudoknots fold into knot-shaped three-dimensional conformations but are not true topological knot

## Uniprot swisprot

Ans:

- Protein Sequences are reported from sequencing experiments
- This data is stored in protein sequence databases
- The famous ones include UniProt & SwissProt

## Scoringatrix formula 5

$$s(a,b) = \frac{1}{\lambda} \log \frac{p_{ab}}{f_a f_b}$$

$s(a,b)=1/\lambda \log p_{ab} / f_a f_b$

**Pam matric steps5"**

Ans:

Step 1: Align proteins sequences which are 1-PAM unit diverged

Step 2: Let $A_{i,j}$ be the number of times $A_i$ is suabstituted by $A_j$

Step 3: Compute the frequency $f_i$ of amino acid $A_i$

Then, PAM1 = $p_{ij}$ = $\frac{A_{ij}}{\sum_k A_{ik}}$

PAM'n' = ( PAM1 ) $^n$

**STEP OF BLOSUM?**

Ans:

**Step 1:** Eliminate sequences that are identical in x% positions

**Step 2:** Compute observed frequency $f_{i,j}$ of aligned pair $A_i$ to $A_j$. Hence, $f_{i,j}$ becomes the probability of aligning $A_i$ and $A_j$ in the selected blocks.

**Step 3:** Compute $f_i$ which is the frequency of observing $A_i$ in the entire block

$$s_{ij} = \log_2 \frac{f_{ij}}{p_{ij}}, \quad p_{ij} = \begin{cases} f_i f_i & i = j \\ 2 f_i f_j & i \neq j. \end{cases}$$

**How are Bulges Formed?**

Ans:

• Bulges, are formed when a double-stranded region cannot form base pairs perfectly

• Bulges can be asymmetric with varying number of base pairs on one side of the loop

**Name three hydrophobic amino acid?**

Ans: glycine (Gly), alanine (Ala), valine (Val), leucine (Leu), isoleucine (Ile), proline (Pro), phenylalanine (Phe), methionine (Met), and tryptophan (Trp).

**2.rooted and unrooted tree difference?**

Ans: Rooted and Unrooted trees can be used to show phylogenetic relationships between sequences

Rooted tree shows the phylogeny of ancestors

**Phylogenetic trees** take several forms: They can be rooted or **unrooted**, binary or general, and may show, or not show, edge lengths. A rooted**tree** is a **tree** in which one of the nodes is stipulated to be the root, and thus the direction of ancestral relationships is determined

- Rooted trees reflect the most basal ancestor of the tree in question.
- Unrooted trees do not imply a known ancestral root.
-

**dynammic programming to create scoring matrix?r 1 series sequence dia hua tha scoring matrix maloom krny k liy?**

ans: From the large number of possible nucleotide combinations, it is hard to find the optimal one

• Dynamic Programming (DP) helps break the problem into smaller problems

Select a nucleotide from sequence

List all possible complementary position for the selected nucleotide in the complete sequence.

How DP works?

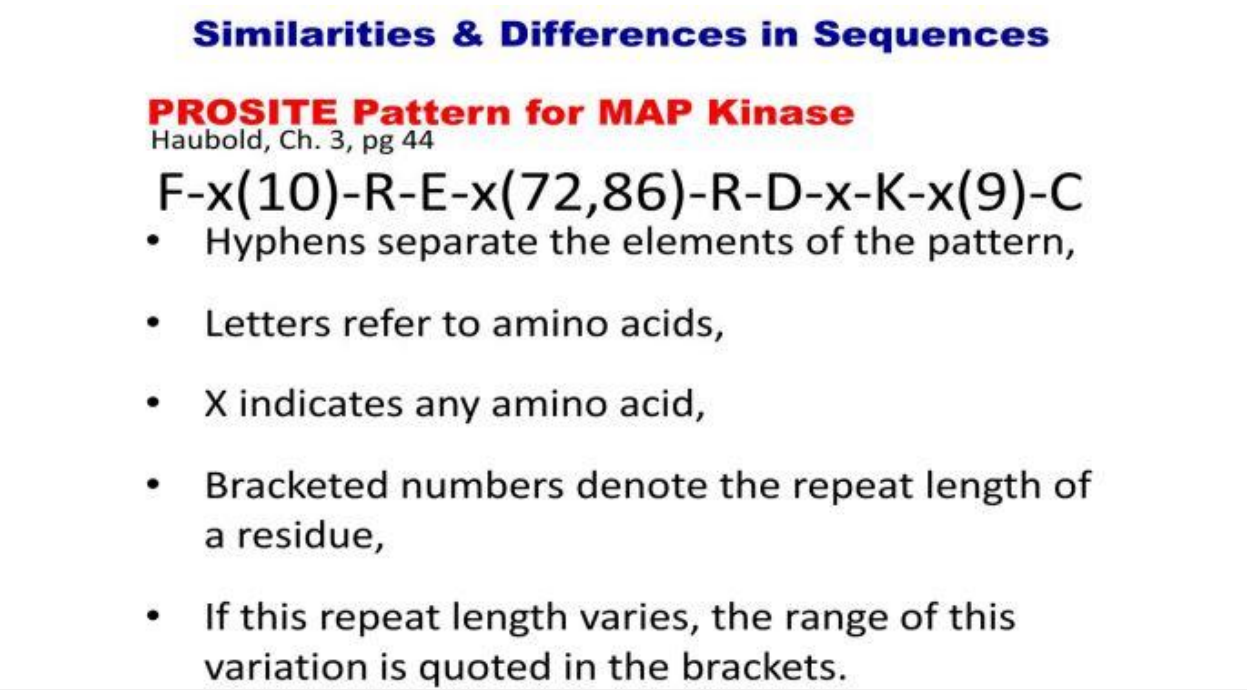
• DP exhaustively makes all possible nucleotide combinations

• DP then recombines such combinations in a process called “Traceback” to ensure that the highest coupled 2’ structure is reported

**Complete the protein sequence .F-x(3)-x-R-F-K-x(4-5)-D-E-R?**

FXXXXRFKXXXXDER

FXXXXRFKXXXXXDER




## Clustering?

***clustering*** is used to build groups of genes with related expression patterns

## Define system biology and its application?

. Systems Biology

**Systems biology** is the computational and mathematical modeling of complex biological systems

Application

• Model protein and gene interactions

• Dynamical analysis of such models

• Understand system properties

• Predict system level behaviors

## Need for bioinformatics ?

1 An interdisciplinary field

2. New but rapidly developing field

3. Low requirement on infrastructure and research equipment

4. Vast opportunities for scientific discovery

Life, evolution and disease can be better understood

• Drugs can be better developed after understanding molecular basis of life

Possible contributions

• Solve important problems related to life

• Handle mammoth quantities of experimental data

• Develop efficient solutions towards these goals

**Role of 5' cap and poly a tail?**

1. The 5' cap helps identify mRNAs at the Ribosomes

2. The 5' cap also acts as a shield against 5' exonuclease, thus leading to an increase in mRNA stability

The 3'-end of mRNA has a polyA tail (around 30-200 adenylate residues) which help shield against 3' exonucleases

# (mid papers year 2017)

STEP OF BLOSUM 5
HOW BULGE ARE FORMED 5
Name three hydrophobic amino acid.
bioinformatic in society role

Pusdoknots?
System biology
Msa

PSA
RNA types
Koi code tha protein name likhnatha

Define system bilogy and its application(2)
What is the need of bioinformatics. 3
Write the methods of clustring 3
Local & global alignments 5
Why RNA has short life span?illustrateur answer with diagram. 5

1.Descride the 3D-ID boeweialgrithim ...2
2.What is the role of 5' cap and poly A tail? ...2
3. What are the methods for clustring the phylogenetic trees.....3
4.write the four objectives of the comparing sequensing?....3

3D-1D boawie algorithm (2)
Purines and pyrimidins structures (2)
Cluster method for construct phylogenetics trees (3)
Global and local alignment differences (3)
One of the method for construction a phylogenetics tree(5)

benefits of dot plot. 2 marks.. EMBL and EBI stands for what, 2 marks.. Differentiate b/w fingerprinting and shotgun...3 marks.. Different field of expasy.. 5 marks Diffetiate between local and global alignment

y hairpin loop reverse the chemical directn of rna structure...2 marks how can we modify the dot plot to consider the gap n mismatch .....2 marks. diffrnc of blast n fasta....3 mrkschallngs of bif.....3 marks bulge formatn....5 prodite pattern diyathaaikusycompltkrnatha...5 marks

Q1: What are Substitutions & Indel? (2) Q2: Why unpaired nucleotides should make the structure of RNA destabilized? (2) Q3: Write formula for finding distance between two sequences and elaborate it. (3) Q4: Top down proteomics. (3) Q5: Write down Four Steps of FASTA. (5) Q6: Differentiate Acidic and Basic Amino Acids. (5) 99% MCQ's were easy; exact from PPTs covering all topics.

1)need of bioinformatics
2)how to make cluster phylogenetic tree
3) difference between global and local alignment

4) progressive alignment
5)nuiseen algorithm

Acronym of blast 2
Most commonly used tools for protien sequence and dna sequence 2
3 differences of alignment sequence and msa 3 marks
Gibbs free energy and its role in rna formation 3
Four positions of NJ algorithm 5

1.Difference b/w UniProt& EMBL?
2. About amino acids and protein which has 3 codons?
3. Scoring Matrix & NJ algorithm?
4. About bioinformatics tools?
5. What is the use of blast & its performance?
6. How Gibbs energy fold Rna Secondary structure?
MCQs were some easy

Usage of sequence data 2 marks
Diff b/w DNA nd RNA on the basis of nucleotide bases. 2 marks
Challanges in the field of bioinformatics.
3 marks
What is Dot plot explain the usage of gap in the dot plot.3 marks
Explain Blosum matrix steps.5 marks
Give diff b/w Rooted nd unrooted also write its uses 5 marks

Describe the steps of PAM?(3)
Hairpin structure of RNA? (2)
Define top down proteomics?(2)
Application of bioinformatics? (3)
Phylogenetics tree types (5)?
Describe system biology and its application? (5)

.....1)central dogma...2marks.....

2)rooted or un rooted tree.....5marks....

3)codon and retendency of codon nad name the amino acid which code three codons....5 marks.....

4),MS proteomics protocol....3 maks.....

5)how RNA structure bntahai from ajeebsanaamtha us sai ...2marks.....

6)pairwise and multiple sequence alignment mai difference any three....3marks
;;;;;;;;;;;;;;;;mcqsbacho 5 ya 6 quiz walye thy or bakikaffi easy thy or
FASTA,BLAST,MS,BLOSUM wgairasai thy or RNA or DNA sai thy

Bif401
meant of substitution and indels
Difference between acidic and basic amino
acid
Difference between top and down proteomics
Write four scoring matrix
Nucleotide meant eloborate

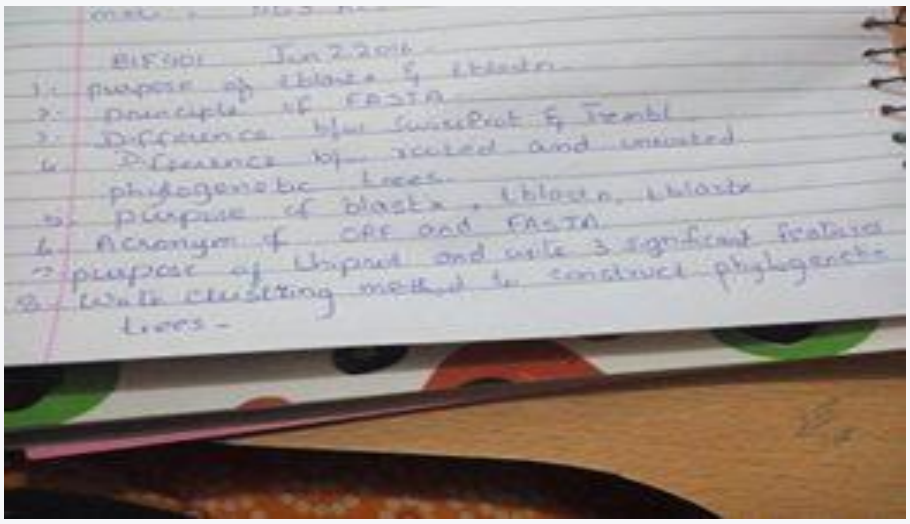
## (Papers before year 2017)

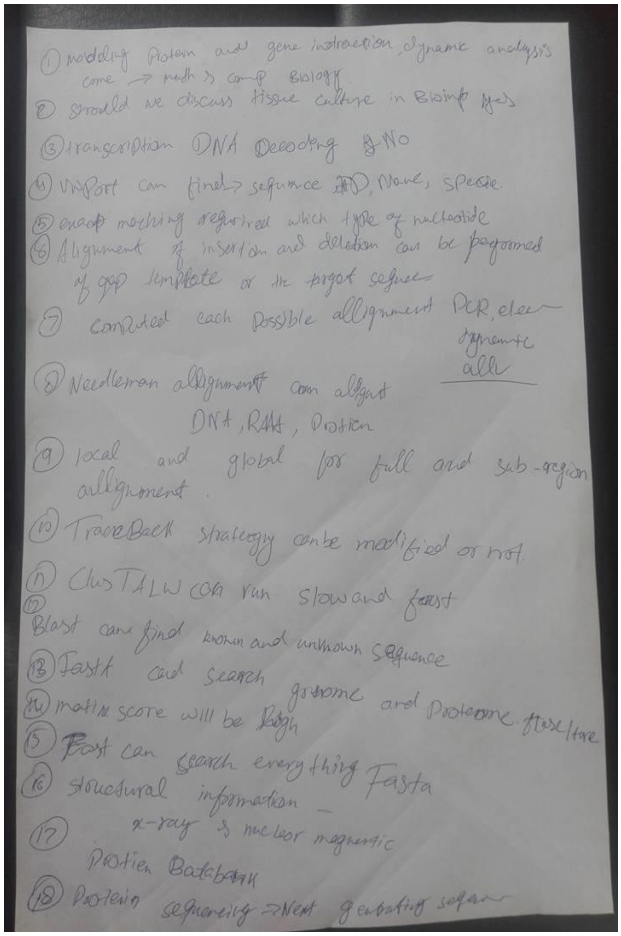
Q1:What are pseudoknots?
Q2:What is the purpose of 3D-1D bowie algorithm
Q3:What is needs of bioinformatics?
Q4:uses of blastx, tblastx and tblastn?
Q5:Steps involved in FASTA algorithm?
Q6:Difference between acidic and basic amino acids?

7 Computed Each possible alignment
PCR, Electrophoresis and Dynamic programing (All)
8 DNA, RNA and Protein sequences can be aligned using Needleman Wunsh algorithm.
9 Local alignment focuses on the sub regions within the sequences. Global alignment can full length
10 Modification in traceback strategy can help find overlap matches containing leading or trailing sequence (yes)
11 CLUSTALW can run
Fast and slow modes
12 BLAST can search sequence databases and identify unknown sequences by comparing them to the known sequences
13 Fasta can search (yes)
14 Matrix score will be high 3rd option tha.
15 Multiple types of FASTA exist which assist in aligning DNA/RNA/Protein sequences (Fasta)

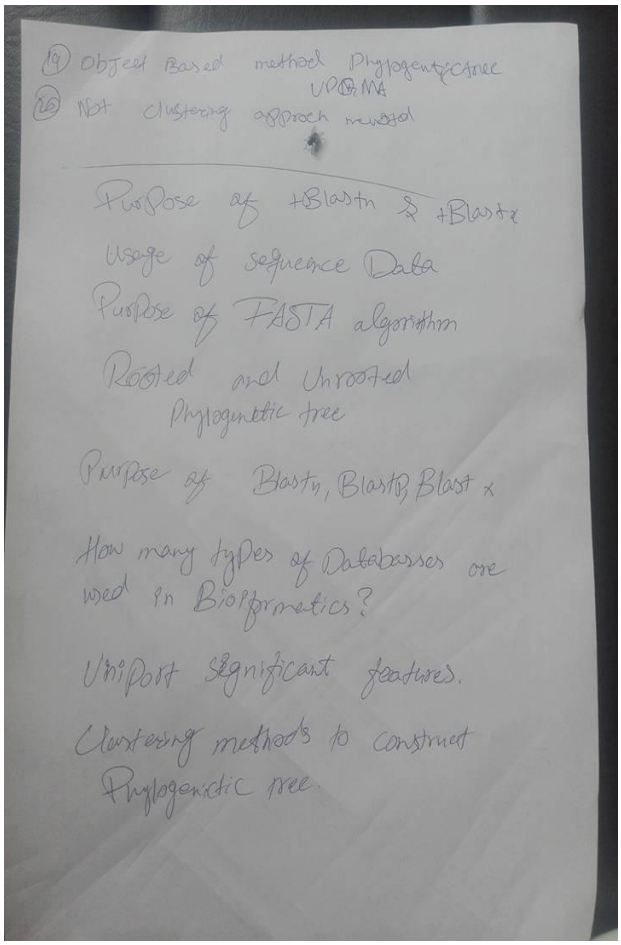
16: Structural information can be get by X-ray Crystallography and NMR
17 Protien Databank See ppt 73
18 Protien Sequencing can be done by
Next generation Sequence
19 Object based method is
Least Square Distances
20 Not Clusturing Approch
Other than
UPGMA
MPGMA
Nieghbor Joining
Single Linkage
Complete Linkage

Saturday 4th June, 2016
BIF401
MCQ's
Identification of low abundance……. in patient tissue samples
DNA is the blue prints of building cells..??? Yes or no
The amino acids polymerize to form:
…….are involved in directing cellular responses to a diverse array of stimuli
Global and Local alignment is used depending on the required type of:
Similar regions are represented by a diagonal row of:
Question:
Difference between Local and Global Alignment
Acronym of FASTA & ORF
Difference between FASTA and BLAST





1. modeling Protein and gene interaction dynamic analysis core → math is comp Biology.
2. Should we discuss tissue culture in Bioinf labs
3. transcription DNA Decoding & NO
4. UniPort can find → sequence ID, Name, species
5. exact matching algorithm which type of nucleotide
6. Alignment if insertion and deletion can be programmed of gap template or the target sequence
7. computed each possible alignment PCR, elec — dynamic alli
8. Needleman alignment can align DNA, RNA, Protein
9. local and global for full and sub-region alignment.
10. TraceBack strategy can be modified or not.
11. ClusTALW can run slow and fastst
12. Blast can find known and unknown sequence
13. FastA can search genome and Proteome structure
14. matrix score will be high
15. Fast can search everything Fasta
16. structural information — x-ray & nuclear magnetic
17. Protien Database
18. Protein sequencing → Next generating sequencing



19. Object Based method Phylogenetic tree UPGMA
20. Not clustering approach needed

Purpose of +Blastn & +Blastx

Usage of sequence Data

Purpose of FASTA algorithm

Rooted and Unrooted Phylogenetic tree

Purpose of Blastn, Blastp, Blast x

How many types of Databasses are used in Bioinformatics?

UniPort significant features.

Clustering methods to construct Phylogenictic tree.